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ARTICLE

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Determinants of football players' valuation: A systematic review

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Abstract

As a result of the growth of the football players' transfer market and the increased availability of data, scholars from different fields have been investigating the topic of the empirical determination of the football players' valuation, that is, transfer fees or monetary valuation proposed by media, for 30 years. We propose a systematic review of the topic to provide the research community with an overview of the determinants of football players' valuation. Peer-reviewed papers written in English on men's football presenting fully available and interpretable econometrical results were selected. Based on the careful analysis of the 29 selected papers and their empirical models, this study has two main implications. First, it classifies all the independent variables used in the literature to explain or predict football players' valuation, and proposes a set of core explanatory variables, with associated expected signs, that constitutes the common ground on which to build further research. Second, by presenting the state of the art in the current empirical literature, it highlights the existing gaps and suggests precise future research objectives. From a more general perspective, the review provides empirical insights

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for researchers interested in the valuation of human resources, signaling theory, market design, or bargaining theory.

KEYWORDS
football players' market value, systematic review, transfer fees, transfer market, valuation

1 | INTRODUCTION

Football players and their employment contracts constitute the main assets of professional football clubs. Hence, the question of their valuation is of great interest for all parties (clubs, players, intermediaries) who need to properly assess what the players are worth to efficiently engage in the transfer market. The interest and the importance of the topic of football players' valuation have increased jointly with the growth of the transfer market which took place after the Bosman ruling by the Court of Justice of the European Union in 1995 (Union Royale Belges Des Sociétés de Football Association and Others v Bosman and Others, 1995). Concurrently, scholars from different fields have discussed the topic of the empirical determination of the football players' valuation for almost 30 years, leading to a vast and heterogeneous landscape of literature characterized by a variety of approaches and methods. This literature provides interesting takeaways for practitioners and football-focused researchers but also for general economists keen on finding empirical tests for economic theories such as Nash's bargaining theory or Spence's signaling theory among others.

In an industry where players sign fixed-term contracts, the Bosman ruling facilitated players' freedom of movement, releasing them from all obligations toward their clubs when their employment contract expires. Combined with the liberalization of the broadcasting markets in several countries which poured money into the system, this led to an increased number of players still under contract being transferred, generating fees paid by buying clubs to compensate former employers. The Bosman ruling was in fact the cornerstone of deregulation of the labor market of professional football players which led to an economic boom (Brocard & Lepetit, 2018). In 2019, 18,080 players were transferred from one team to another based in a different country, which generated a total of 7.35 billion dollars paid by clubs to recruit players protected by employment contracts (FIFA TMS, 2020)[1].

Following Bosman, researchers acknowledged that the services provided by football players should be recognized as accounting assets for clubs (Morrow, 1996). From this point on, players are considered depreciable intangible assets when clubs invest in their recruitment, that is, pay a transfer fee to recruit them (e.g., in France see (Règlement n°2004-07, 2004)). As football clubs' businesses rely more heavily on labor than on capital, players are core assets for clubs and their valuation becomes an important determinant of clubs' financial value (Forker, 2005; Scelles et al., 2016).

The valuation of a player, like any valuation—such as the valuation of a company—can have several purposes (Matschke & Brösel, 2021). The first goal could be to value the assets of the club within the annual financial statement (Morrow, 1997; Oprean & Oprisor, 2014; Pavlovic et al., 2014), but is practically performed at various points by several stakeholders. Indeed, when clubs

are faced with the decision to recruit a player still under contract with another club, this should be considered as an investment. The employer club, when contacted by a club interested in acquiring the services of a player under contract, is expected to set an asking price. In parallel, if a player has an expiring contract, the employer club needs to consider whether to extend the contract or whether it makes sense to channel the freed-up capital into an alternative investment. Therefore, valuation in general and the valuation of a player from the club's perspective takes place for decision-making purposes (Leifheit & Follert, 2021). Moreover, in a widely intermediated market (Brocard & Cavagnac, 2017), it is also interesting and helpful for the players' representatives to assess what value the club attaches to the athlete when they enter into a contract negotiation about his salary. The more valuable the player, the higher the salary his representatives will be able to negotiate.

Two types of players' monetary valuations are available to researchers. First, they can refer to transfer fees actually paid by clubs, which represent the price of a specific player at a certain time, as set by buyers and sellers in the market. Unfortunately, the actual transfer fees are very rarely publicly disclosed by the contracting parties and are only made available by specialized newspapers and websites. The same kind of issues exists for contractual clauses such as salaries, add-ons (payments conditional on player's performance or other agreed terms), or sell-on (an agreed amount or percentage to be paid by the buying club to the current club if it obtains a transfer fee for the player in the future). Second, media companies, such as Transfermarkt.de, Marca or Kicker.de provides figures called "market values" which are branded as a valuation available for every player at every moment either based on experts' judgment or on the suggestion and rationale of every member of a community (critically to this concept, Ackermann and Follert (2018)). Continuously offering updated valuations to every player, Transfermarkt values have notably gained interest among football fans and scholars working on the transfer market. This valuation, which validity is mainly based on the concept of the wisdom of the crowds (Surowiecki, 2005), aims at offering a (subjective) monetary value to every player by gathering and merging registered fans' opinions. According to the website, these crowd-sourced valuations do noy aim at predicting a price, that is, the transfer fee, but "an expected value of a player in a free market" (*Transfermarkt Market Value Explained*, 2021). Therefore, the value estimated by the community and the price actually observed when the transfer materializes have important conceptual differences. However, researchers have acknowledged their empirical proximity (Coates & Parshakov, 2022; He et al., 2015; Herm et al., 2014; Müller et al., 2017; Rubio Martín et al., 2022; Serna Rodríguez et al., 2019) which leads us, in this article, to regroup these two concepts under the umbrella of "players' valuation" and focus on the determinants of such a valuation. Social science research makes a valuable contribution here by empirically investigating these determinants.

The aim of this paper is to shed light on the empirical approaches used to determine the influencing parameters of players' valuation. To this end, we use a systematic literature review with a special focus on several quantitative elements complying with PRISMA guidelines (Page et al., 2021). We believe it is urgent to conduct a systematic review in the face of the abundance of literature, especially because the field is worked on by scholars from different disciplines so that the results can be found in journals specializing in Economics, Business Economics, Sports Science, Artificial Intelligence, and other fields. A meta-analysis was considered but subsampling to keep only articles focusing on a specific dependent variable, that is, a specific valuation, would have led to a limited articles sample and hardly generalizable findings. Therefore, we preferably consider several types of valuation to propose a broader systematic review with quantitative elements. If systematic literature reviews are more and more frequently used in economics and management (Clark et al., 2021; Fisch & Block, 2018), to the best of our knowledge, on our specific matter, only

one systematic review has been conducted but represented only a preliminary step in the analysis developed in the article (Serna Rodríguez et al., 2019). It is worth noting that fewer databases were searched, and some terms not queried so that our sample of articles appears to be more comprehensive. Also, one peer-reviewed article had the precise purpose to summarize the state of the art at the time of publication but no systematic method was used (Frick, 2007).

The main research objective of our systematic literature review is to determine and classify the crucial drivers of the football players' valuation examined by empirical methods. In addition, the reader should bear in mind that subsequent goals are to present the characteristics of the empirical literature on football players' valuation, detail the theoretical frameworks called upon, provide actionable items for future research, and highlight the gaps the latter would need to fill in.

The remainder of the paper is structured as follows. The second section details our systematic review methodology that helps understand and evaluate the results presented in the third section. These results describe the dependent variables, the econometric techniques, the populations studied, and the independent variables used in the empirical literature. In the fourth section, we discuss our main results, the pitfalls to avoid in future work, and the limitations of our approach, before providing the reader with a set of key takeaways and proposing six future research objectives for the field in the conclusive fifth section.

2 | METHOD

The method of the systematic review was chosen to answer the research question of the present study. The detailed systematic review strategy complying with PRISMA guidelines is available in the Online Appendix A, but this section presents its salient facts.

On November 19th, 2021, to find all articles related to the players' valuation topic we searched the terms "football player value," "football player price," "football transfer fee," and several other variations on multiple multidisciplinary databases. Successively screening the titles and abstracts of the collected papers, and assessing their eligibility given our research framework, we filtered 3298 first identified papers to keep the 29 articles which results are presented hereafter. A summarizing table of the 29 selected articles and their main empirical characteristics is available in the Online Appendix B. The selected articles were written in English, the language of almost all the research published on the topic and commonly shared by the four authors, published in peer-reviewed journals, to ensure a minimal quality standard, and focused on men's football. To assess the relevance of the papers, we only selected the ones presenting empirical models in which the dependent variable was the observed transfer fee or another publicly available and reliable monetary valuation of football players (Transfermarkt values, Marca values, or kicker.de values). These articles needed to present coherent econometric models allowing for a relatively consistent comparison of the results. Therefore, to ensure that models are interpretable and comparable, we considered only econometrical models presenting explainable coefficients along with p-values or t-stats, thus excluding black box methods or insufficiently detailed results.

Once selected, each one of the 29 eligible articles was read independently by two authors and every reader filled a jointly validated reading sheet. This reading sheet contained information regarding the field of publication, whether the focus of the article is the valuation of the football players, the research question, the research results, the interest for our review, the value or price definitions used, and the future research ideas. Finally, every included article can showcase several model specifications, leading us to a total of 111 specifications and an average of 3.83

specifications per article. The empirical results were compiled in a separate spreadsheet (publicly available online) to allow the authors to produce the qualitative and quantitative analysis presented hereafter.

3 | RESULTS

Our literature review provides a comprehensive view of the independent variables tested in the empirical analysis of football players' valuation. The following subsection introduces these results by specifying the empirical research context in which the diverse independent variables were used. The dependent variables explained, the modeling approaches implemented as well as the population studied were investigated. The second subsection then breaks down these results into a classification of the variables, an overview of the theoretical justifications, and a systematic analysis of the main explanatory variables and control variables used in the literature.

3.1 | Empirical research context

3.1.1 | Dependent variables

Before examining the details of the model specifications in our sample, it is instructive to define the different dependent variables used, to discuss their validity and their distribution in our sample. First and foremost, 13 articles propose at least one model specification where the dependent variable is the transfer fee. A transfer fee is the amount of money that a buying club pays to obtain the early termination of the contract of the coveted player from the current club. Therefore, the transfer fee is often considered the value of the services of the player. Also, two articles used arbitrated fees as dependent variable. In the late 70s, 80s, and early 90s, under precise circumstances, the English clubs could appeal to the Football League Appeals Committee to arbitrate the fee amount due from the buying club to the selling club instead of using the usual negotiation way (Speight & Thomas, 1997b; Wood, 1988). The system fell into disuse and the research on arbitrated fees seem to be neglected in the economics and management literatures nowadays. Interestingly, one article in our sample shows that there are no strong contradictory results between arbitrated and negotiated fees even if arbitrated ones are significantly lower and are less sensitive to clubs' characteristics, that is, bargaining strength (Speight & Thomas, 1997c). The transfer fee, either negotiated or arbitrated, is an objective and (theoretically) observable information. The associated downsize is that it is not available for every player at any time but rather only when a transfer eventually occurs, leading to selection bias[2] (Carmichael et al., 1999; Dimitropoulos et al., 2018; Ruijg & van Ophem, 2015).

This situation may explain why 15 articles in our sample rely on crowdsourced estimates of players value, retrieved from the website Transfermarkt, as the dependent variable in at least one of their model specifications. The broad, easy and continuous availability of the Transfermarkt valuation makes it possible for interested researchers to study the determinants of the crowdsourced valuation and avoid the selection bias. Over the last years, the validity of the Transfermarkt crowdsourced valuation has been tested by many articles focusing on their ability to be used as proxies for transfer fees (Coates & Parshakov, 2022; He et al., 2015; Herm et al., 2014; Müller et al., 2017; Rubio Martín et al., 2022) or for salaries (Prockl & Frick, 2018; Thrane, 2019; Torgler & Schmidt, 2007). Crowdsourced estimates are now considered to be good proxies for transfer fees even if they

tend to underestimate transfer fees (Coates & Parshakov, 2022) and to be more accurate than transfer fees estimation models for the higher segment of the value hierarchy (Müller et al., 2017). Also, two articles use subjective valuations proposed by other media companies, Kicker.de and Marca. The former has been considered a good source for reliable players' valuation (Buraimo et al., 2015; Frick, 2011). There is no evidence that the latter can be used as a good proxy, but we decided to include the paper using it in our sample as it was published when studies on players' valuation where very scarce, relied on a solid theoretical basis and was the first to consider popularity as a predictor for valuation (Garcia-del-Barrio & Pujol, 2007).

3.1.2 | Modelling approaches

As previously mentioned, we intentionally focused on model specifications allowing inter-article comparison and interpretation, thus restricting the analysis to methods presenting beta coefficients and associated *p*-values or *t*-stats. Out of our 111 models, the most frequent method was Ordinary Least Squares with 94 specifications, many of which used panel regression with fixed or random effects. Some other methods were used to correct biases or non-endogeneity issues:

- Generalized Method of Moments (seven specifications) was used to handle endogeneity between the valuation and appearances of foreign players (Dimitropoulos et al., 2018),
- 95% quantile OLS regression (3), was used to study the higher segment of the transfer market and the superstar effect (Franck & Nüesch, 2012; Herm et al., 2014),
- Feasible Generalized Least Squares or Generalized Least Squares (3), were used to handle heteroscedasticity (Majewski, 2016, 2021),
- Tobit models (3), were used to handle selection bias in the transfer fees, that is, only transferred players fees are available while we do not know the dependent variable for players not transferred for a fee (Carmichael et al., 1999; Ruijg & van Ophem, 2015),
- Two-stage least squares (1), were used to handle endogeneity issues (Serna Rodríguez et al., 2019).

3.1.3 | Populations studied

Obviously, over a 30-year span the population studied by the researchers are incredibly heterogeneous. The seasons (Figure 1) and leagues (Figure 2) considered make the results of the different specifications hard to compare. Therefore, we present the general characteristics of the specifications of our sample to have a better overall picture of the empirical literature produced until now and to show that this heterogeneity produces a point about the generality of results and their ability to be observed in a wide variety of contexts. Following the distribution of the years of publication, it is quite logical to see that most of the articles and specifications focus on seasons in the early 2010s. The number of specifications considering a season range from 1 article for the 1978–1979, 1979–1980, 1980–1981, 1981–1982, 1982–1983, 1983–1984, 1984–1985, 1997–1998, 1998–1999, 1999–2000, 2000–2001 seasons to 11 articles for the 2011–2012 season (or 2011 depending on the league considered). The results for the number of specifications produced for every season is quite similar, ranging from two specifications for the 1978–1979, 1979–1980, 1980–1981, 1981–1982, 1982–1983, 1983–1984, 1984–1985 seasons to 51 specifications built on 2012–2013 or 2012 data.

Regarding the leagues, the English leagues are very well represented as the absence of articles focusing on other leagues until the late 2000s illustrates. The Premier League (57 specifications,

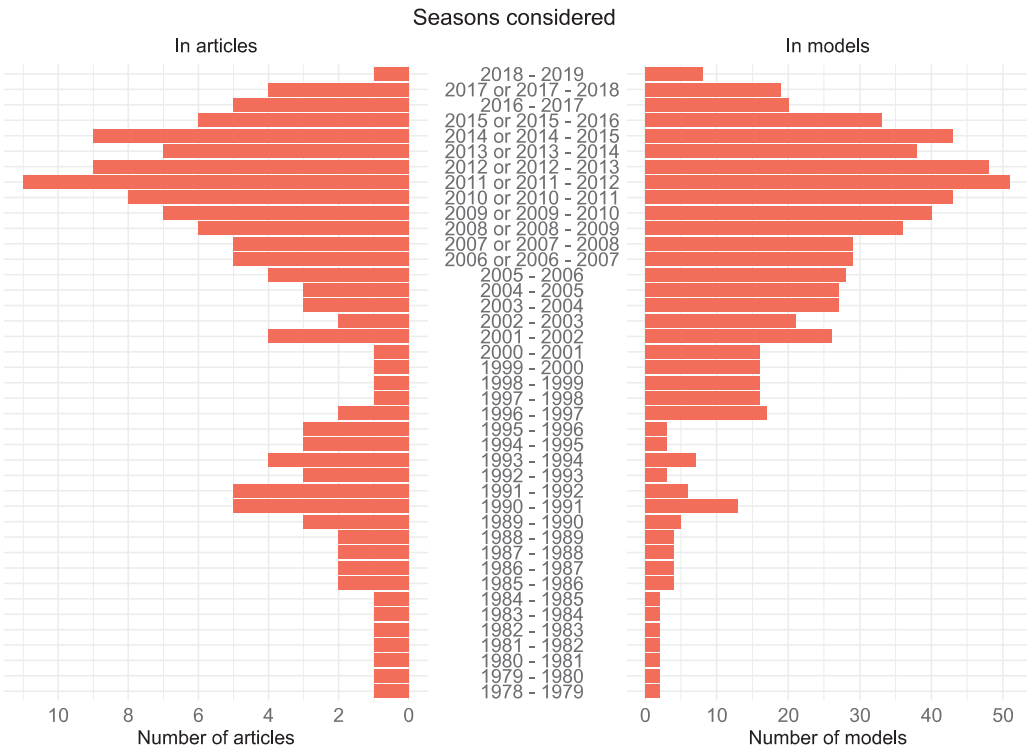

FIGURE 1 Number of model specifications and number of articles by season. [Colour figure can be viewed at wileyonlinelibrary.com]

16 articles) is the most studied league followed by other "big-5" leagues: German Bundesliga (38 specifications, nine articles), Spanish Liga (31 specifications, seven articles), French Ligue 1 (22 specifications, five articles), and Italian Serie A (22 specifications, five articles). Given the availability of English data in the 90s, the lower English divisions are also surprisingly well represented in our sample, the Championship (25 specifications, 8 articles) being more tested than Ligue 1 and Serie A for instance. Aside from English and Big-5 leagues, none have been studied by more than one article, leading to a very wide sample of leagues studied, mainly European ones with two exceptions: the American Major League Soccer (Prockl & Frick, 2018) and the Indonesian Liga 1 (Adiwiyana & Harymawan, 2021).

The details of the modelling techniques used, sample sizes, number of specifications included, leagues and seasons considered, data sources, fixed and random effects included for every article are provided in the Online Appendix B.

3.2 | Independent variables

3.2.1 | A six-category classification of independent variables

To summarize three decades of research and multiple model specifications, we attempted to list all the independent variables that have been introduced in model specifications and to classify them by conceptual proximity on three different levels (Figure 3).

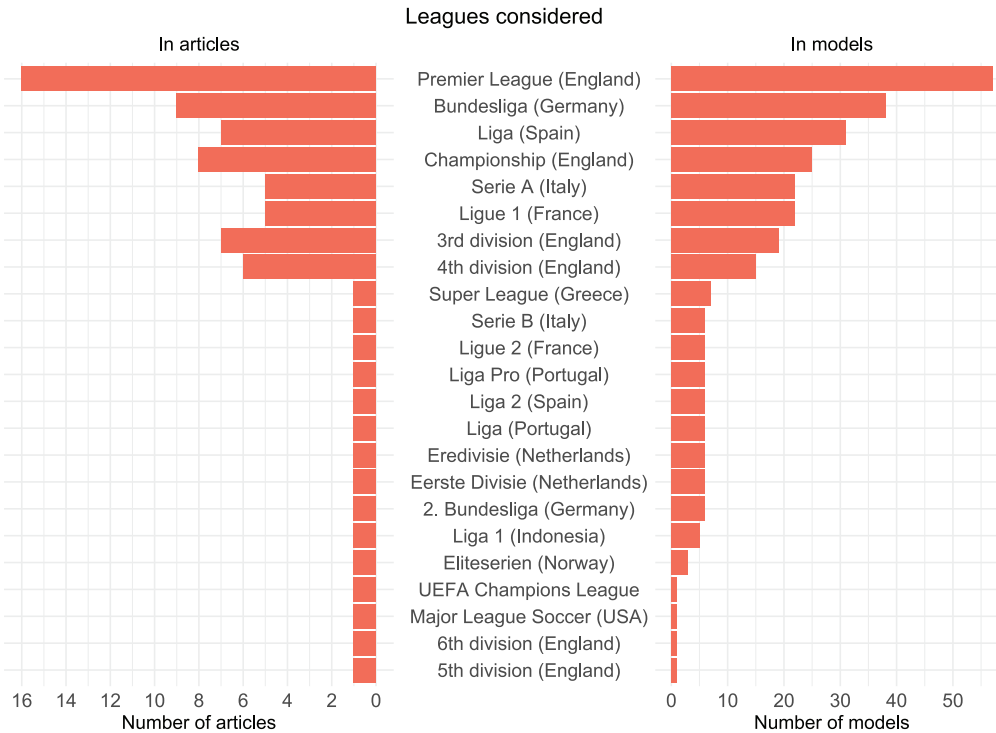

FIGURE 2 Number of model specifications and number of articles by league. [Colour figure can be viewed at wileyonlinelibrary.com]

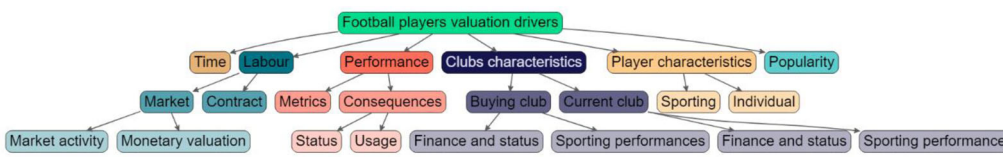

FIGURE 3 Classification of the variables included in the literature. [Colour figure can be viewed at wileyonlinelibrary.com]

We compiled 471 variables included in all the articles of our sample and regrouped them into 93 generic variables, split into six categories; themselves broken down into subcategories when deemed relevant (Online Appendix C):

- Clubs' characteristics clusters both current and buying clubs' characteristics. These characteristics are split between financial characteristics (e.g., *profit, ownership,* or *revenues*), status (e.g., *attendance* or number of foreigners), and sporting performance (e.g., *goal difference, league ranking,* or *coaches*). Articles including buying clubs' characteristics are rarer. The main reason is that buying clubs' characteristics are only relevant when considering transfer fee as dependent variable and no other monetary valuation (Kicker, Marca, Transfermarkt).

- Time refers to only three variables included in the models to consider the effects of passing time (*seasons, years*, and *months*). These variables are usually considered fixed effects.
- Labor regroups variables dealing with the employment contract of the player (*clauses*, *contract length*), variables describing the market design (e.g., *arbitration, transfer window,* or *clubs power balance*) and other valuations (*previous transfer fee, Transfermarkt value*).
- Performance gathers both variables measuring the performance of the player (e.g., *goals, assists,* or *saves*) and the direct or indirect consequences of this performance (e.g., *minutes played, journal ratings,* or *trophies*).
- Player's characteristics includes variables related to his individual characteristics (e.g., *height, age,* or *nationality*) but also to his sporting career (e.g., *seniority, number of clubs,* or *youth academy*).
- Popularity is measured by *social media activity* (e.g., Instagram, Reddit, YouTube), *press citations* or *internet searches* (mainly Google).

Performance appears to be the most studied category. Indeed, almost all articles include performance metrics and some of them even test several advanced metrics (Müller et al., 2017). *Age, position, nationality,* or *footedness* are also very commonly tested and describe player's characteristics in our categorization. Clubs' characteristics have also been tested by many researchers in the early literature to assess market powers of current and buying clubs (Carmichael & Thomas, 1993; Gerrard & Dobson, 2000; Speight & Thomas, 1997c). Labor, Popularity and Time are more rarely included in analyzed articles. To have a more in-depth view of the determinants tested in the literature our presentation of results is threefold. First, we draw up the contours of the theoretical models discussed in the literature to summarize how they shaped the econometrical design and the inclusion of certain variables. We then assess the degree of empirical validation of the theoretical models and highlight that the results provide evidence to support economic theory regarding bargaining theory, market structure, and signaling theory. Second, to summarize the empirical results of the literature dedicated to valuation, we analyze the 10 most-tested variables in a systematic and quantitative way to extract a set of core variables built upon the accumulation of findings from the selected articles. Finally, we conclude this overview of the determinants of football players' valuation by presenting the random and fixed effects used in the literature to control for interpopulation heterogeneity.

3.2.2 | The confrontation of theoretical models with empirical results

Several pieces of literature in the field developed strong theoretical frameworks to characterize the transfer market and consequently studied the football players' valuation. However, we must underline that these theoretically-backed results are mainly the work of early researchers, and that more recently, the field has slightly shifted toward valuation per se and the validity of the Transfermarkt valuation.

First, calling on Nash's bargaining theory (1950, 1953), Carmichael and Thomas (1993) advocate that clubs must engage in a negotiation to transfer a player; hence the final fee is the result of the confrontation of clubs' expected utility functions and their status quo utility. Therefore, clubs' characteristics are expected to influence their bargaining power and are a priori considered potential explanatory variables of the transfer fee. It is arguable that the literature provides important insights to the bargaining strengths of the selling and buying clubs by testing clubs-related variables (Carmichael & Thomas, 1993; Dobson et al., 2000; Gerrard, 2001; Gerrard & Dobson, 2000;

Reilly & Witt, 1995; Rubio Martín et al., 2022; Speight & Thomas, 1997a, 1997c). On the one hand, for the buying club, the *attendance* has a positive and significant effect on transfer fee, just as the *goal difference*, the *UEFA points*, or the *division in English leagues*. For *buying club's profit* and *stadium capacity* there is no evidence of a clear effect. On the other hand, for the current club, only the *divisional status* has a quite reasonable level of evidence of being positively and significantly impacting the player valuation. Concerning the current club's *UEFA points, goal difference, attendance*, and *profit*, no strong evidence have been found. Finally, *current clubs' rankings* in the league table is subject to mixed evidence (Adiwiyana & Harymawan, 2021; Dimitropoulos et al., 2018; Majewski, 2016; Velema, 2019). To summarize these results, we can say that the buying club's bargaining power seems to show higher evidence of influencing the valuation of player than the selling club's power.

Second, studying the valuation of players provides a perfect empirical setup to test for the existence of monopsony and monopoly rents for clubs vis-à-vis the players or other clubs. According to Gerrard and Dobson (2000), the abovementioned results regarding the clubs-related variables might actually indicate the presence of a monopoly rents component in the transfer fees, meaning that the current club extracts a share of the gap between its reservation price and the transfer fee. This potential monopoly rent was already suggested—but not tested—by Speight and Thomas (1997a) along with the existence of a monopsony rent potentially yielded by the buying club when the current club is in urgent seek of money, needs to sell and cannot search for multiple buyers. If we consider the evidence regarding the importance of clubs' characteristics in the determination of player valuation, it is surprising that they have not been included in more recent studies. In another article with strong theoretical basis, Garcia-del-Barrio and Pujol (2007) suggest that clubs can obtain monopsony rents from players with limited bargaining power that are then forwarded to superstar players. Consequently, the market for players could be considered a "winner-take-all market," that is, a market in which a slight improvement in performance yield much larger increase in economic returns. To support this theory, they show that the popularity of players—proxied by the *number of google citations*—has a positive and significant impact on valuation. This result has been reproduced since with the *number of Instagram followers* (Rubio Martín et al., 2022) and *Google or press citations* (Franck & Nüesch, 2012; Herm et al., 2014), but cannot be confirmed when the variable is log-transformed (Müller et al., 2017; Serna Rodríguez et al., 2019). More generally, according to these authors, the results tend to confirm the segmentation of the player market and the existence of a superstar phenomenon (Adler, 1985; Rosen, 1981).

Third, some more recent studies also tested hypothesis with strong implication for economic scholars, especially those interested in Spence's signaling theory (Spence, 1973). Indeed, Wicker et al. (2013) tested the empirical validity of the idea that effort could be a signaling device on the labor market. However, the authors showed that effort—measured by *intensive runs* and *running distance*—had an insignificant or negative impact on logged valuation and changes in valuation. Velema (2019) studied a different signal by testing the sensitivity of the players' valuation to their upward or downward professional mobility. Job moves are thus conceptualized as signals of their productivity to misinformed potential recruiters. The results tend to show that the *mobility to higher ranked team* leads to a significantly higher valuation.[3] Dimitropoulos et al. (2018) also provide interesting results regarding movements of football players by showing that the impact of the number of games played on valuation is more important for natives than for migrants in the Greek league. Migrants thus need a fewer number of games to attain the same valuation. These results suggest that the status of migrant player is a signal for higher valuation in second-tier leagues.

Fourth, analyzing works which provide sizable evidence for economic theory, we should not miss on the opportunity to highlight that football has also been an early laboratory for the study

of racial discrimination. Indeed, Reilly and Witt (1995) tested the potential "unequal treatment of black players in terms of transfer prices." Their empirical results, replicated by Medcalfe (2008), suggest that there is no such evidence.

Finally, Richau et al. (2021) build on property-rights theory (Demsetz, 1967) and foreign direct investments analysis to study the role of ownership on players' valuation. The role of ownership concentration in valuation, measured by the *investor level of participation*, is non-significant and club's *majority owner origin* shows mixed evidence.

Unfortunately, these studies testing theoretically backed determinants have rarely been replicated, leading to an insufficient level of evidence to be part of the comparative framework presented in the next section. This situation may be due to disappointing results in the first studies leading auto-selection of scholars toward theories with more encouraging empirical validation, or maybe even publication bias. Interestingly, the following most tested variables do not rely on strong theoretical economic basis but rather on pragmatic observation, intuition, and football knowledge.

3.2.3 | The main set of explanatory variables

The number of articles retrieved leads to a very large number of different independent variables with many unique testing and original specification, making it hard to compare results. Therefore, we focus on a subset of the ten most tested continuous or binary variables. We ensure that the way variables were defined in the different articles allowed for sufficient methodological proximity. These 10 variables were used in at least five different articles each. We present a quick description of the analyzed variables any more detail can be found in the online Appendix D:

- *Age* and age squared: Age on the transfer date or on a fixed date identical for every player (sometimes not specified).
- *Minutes played*: The total number of minutes played by the player in the last season.
- *Goals*: Goals scored in the last season.
- *Assists*: The number of seasons assists in the last season.
- *Yellow* and *red cards*: Number of yellow and red cards in the last season
- *Appearances*: Number of matches played by time period. Most of the time it is by season, but it can be by half season (Wicker et al., 2013).
- *National team status*: Binary variable that takes value 1 if the player has participated in senior national team matches.
- *Both feet*: binary (ability to play with both feet or not) or 3-level (left, right, both) variable coded with binaries indicating the ability of the player to play with both feet.

Online Appendix D also presents the availability of results for these variables for the 29 articles in our sample. These definitions are quite restrictive and may explain why, for every considered variable, there are relatively few articles considered. We want to emphasize that if a paper's specific use of a variable is not included, it means that the specification used was not consistent enough to be included and compared with other papers. For example, *goals* and *assists* can be scaled by number of minutes played (Coates & Parshakov, 2022). Also, some articles may use more precise estimated equation, for example, with interactions between *goals* and *position* on the pitch (Carmichael et al., 1999), or more complex variables, for example, using *number of games or minutes played with national team* instead of a binary variable to assess international status

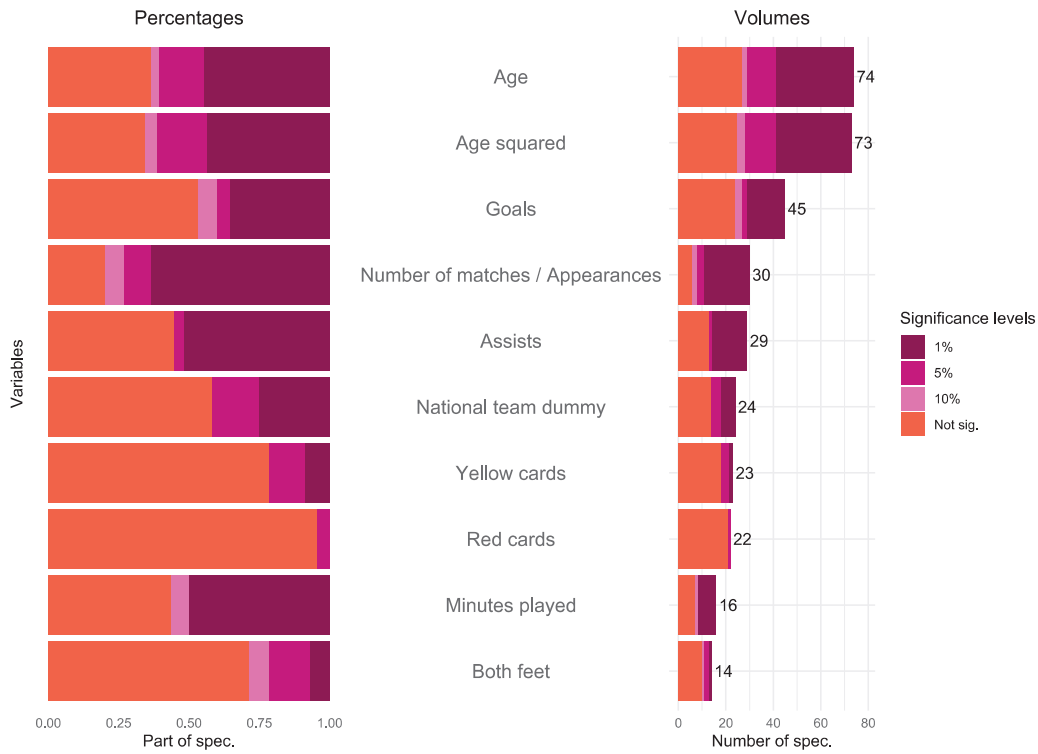

FIGURE 4 Distribution of specifications by significance level for the 10 most-tested continuous or binary variables. [Colour figure can be viewed at wileyonlinelibrary.com]

(Gerrard & Dobson, 2000). Therefore, the fact that the equation estimations can be very specific explains why some articles included in our sample do not present results for any of the 10 variables studied. Garcia-del-Barrio and Pujol (2020) assume that the media visibility encapsulates the player performance, therefore, no performance indicators are used, but only media value and some other "contextual" variables. From our 10 variables of interest, Kalén et al. (2019) only used an *age* variable, but it was stratified thus incomparable. Finally, Majewski (2021) used a parsimonious equation with three variables: *assist* + *goals*, *team valuation* and a binary variable for *brand*. The last preliminary remark on the analysis of the significance and sign of the 10 variables is that the significance levels detailed by every article can be quite different. Seventeen papers display the significance levels at all 1%, 5%, and 10% thresholds. Providing the most precise and helpful results that constitutes the best practice. Unfortunately, 12 papers out of 29 do not detail their results at the 10% threshold. On the other hand, every paper provides significance results at 1% and/or 5% levels. Given this lack of similar standards in the reporting of significance levels, it would be misleading to discuss the results at the 10% threshold (however presented in Figure 4 for comprehensiveness). Therefore, we only present the results at the 1% and 5% levels hereafter.

Figure 5 precises the sign of the relation between the variable and the players' valuation for the model specifications where the considered variable is significant at 5%. The variables with the most empirical evidence are *age* and *age squared* included in 74 and 73 specifications, respectively. The *age* variable is significant at 1% in 33 specifications and at 5% in 45 specifications. Among these 45 specifications, its sign is positive in 32. Regarding the quadratic form, *age squared* is significant at 1% in 32 specifications and at 5% in 45 specifications. At 5%, its sign is negative in 41 of the 45

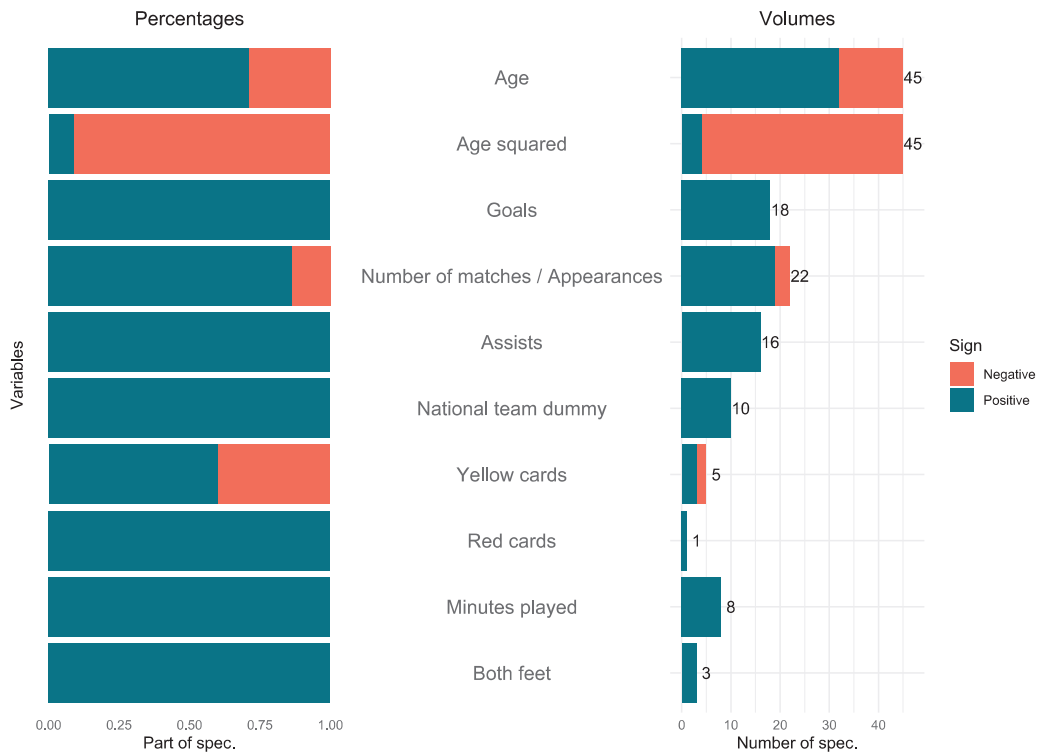

FIGURE 5 Distribution of specifications by sign for the 10 most-tested continuous or binary variables when significant at 5% threshold. [Colour figure can be viewed at wileyonlinelibrary.com]

specifications. More precisely, given the frequent usage of *age* and/or *age squared* in the model specifications and the heterogeneity of results when considering the signs, we analyze the function linking *age* to the valuation of the player. Among the 111 specifications, 29 did not use either *age* or *age squared* in their equation. Regarding the 82 remaining specifications, we observe:

- 23 specifications with non-significant relation (Neither *age* nor *age squared* are significant).
- 44 specifications with inverted-U-shape relation (*age* is positive and significant, and *age squared* is negative and significant).
- Three specifications with U-shape relation (*age* is negative and significant, and *age squared* positive and significant).
- 11 specifications with decreasing relation (*age* is negative and significant, and *age squared* is either not tested or non-significant).

In (Prockl & Frick, 2018), *age cube* is also included. The three coefficients are significant and the one associated to *age* is negative, the one to *age squared* is positive and the one to *age cube* is negative. According to the values of the coefficient, the valuation of the player is thus increasing between 18 and 34 but decreasing after 34.

Another individual characteristic, the capacity to play with *both feet* is the variable with the fewest empirical evidence, being included in "only" 14 specifications. More importantly, the variable is not significant in 10 of these 14 specifications. In the three specifications where the *both feet* variable is significant at 5%, it always has a positive impact on player valuation. The *national*

team variable has been tested in 24 specifications of the sample but has been found to be significant at 1% in six of them and at 5% in 10 of them. Even if these significance results are mixed, we observe that the variable is always associated with a positive sign in the 10 specifications where it is significant at 5%.

Focusing on the performance variables, *goals, number of matches, assists*, *yellow cards*, *red cards*, and *minutes played*, we note that *goals* and *assists*, that is, decisiveness variables, are most of the time conjointly tested just as *yellow* and *red cards*, that is, aggressiveness variables. On the other hand, *minutes played*, and *number of matches*, that is, player's usage variables, are used as mutual substitutes. Regarding the decisiveness variables, out of 45 specifications, *goals* is significant in 16 specifications at 1% and 18 specifications at 5% while *assists* is significant in 15 specifications at 1% and 16 specifications at 5% out of 29 specifications. In the 18 and 16 specifications in which they are significant at 5%, both variables are always positively associated with players' valuation. Concerning the aggressiveness variables, *yellow cards* is included in 23 specifications and *red cards* in 22. These variables are the ones associated with the fewest number and proportion of significant specifications, *yellow cards* are significant in two specifications at 1% and five specifications at 5% while *red cards* is never significant at 1% and only once at 5%. Regarding the sign of the relationship, there is mixed evidence for *yellow card* with the variable being associated with a positive coefficient in three specifications and with a negative coefficient in two specifications. *Red cards* are positively related to valuation in the only model where it is significant at 5%. In terms of player's usage variables, *number of matches* and *minutes played* were included in 30 and 16 specifications respectively, being significant at 1% in 19 and eight of them and significant at 5% in 22 and eight of them. At 5% threshold, all eight significant coefficients associated with *minutes played* are positive. For the *number of matches*, out of 22 specifications where the variable is significant at 5%, the coefficients are positive in 19 and negative in three.

From an outsider viewpoint, the variables studied here in a systematic way could be considered to have mixed results. However, this lack of significance in certain specifications is often due to the role of interactions between variables. In this regard, the analysis of the fixed and random effects included in the model specifications is crucial.

3.2.4 | Other control variables

Finally, to complete the overview of the main control variables useful for future empirical setups, we analyze the categorical variables introduced in the model specifications. Indeed, some variables are quite common but categorical, preventing us from including them in the systematic analysis presented above. Several fixed or random effects are considered in the regressions analyzed: *seasons* (Dimitropoulos et al., 2018; Dobson et al., 2000; Garcia-del-Barrio & Pujol, 2020; Gerrard & Dobson, 2000; Müller et al., 2017; Prockl & Frick, 2018; Richau et al., 2019; Velema, 2019), *current clubs* (Dimitropoulos et al., 2018; Dobson et al., 2000; Franck & Nüesch, 2012; Garcia-del-Barrio & Pujol, 2007; Prockl & Frick, 2018; Thrane, 2019; Wicker et al., 2013), *current club's countries* (Coates & Parshakov, 2022; Velema, 2019), *buying club's countries* (Coates & Parshakov, 2022), *leagues* (Dobson et al., 2000; Müller et al., 2017; Rubio Martín et al., 2022; Serna Rodríguez et al., 2019), and *positions* (Carmichael et al., 1999; Carmichael & Thomas, 1993; Coates & Parshakov, 2022; Franck & Nüesch, 2012; Garcia-del-Barrio & Pujol, 2007; Kalén et al., 2019; Müller et al., 2017; Prockl & Frick, 2018; Richau et al., 2021; Rubio Martín et al., 2022; Ruijg & van Ophem, 2015; Serna Rodríguez et al., 2019; Speight & Thomas, 1997a, 1997c; Wicker et al., 2013). Unfortunately, the high number of categories for *seasons*, *countries* and *clubs* makes it hard

for researchers to publish the detailed results of their regressions. For *leagues* and *positions*, the coefficients and significance levels are sometimes detailed in the regression results. In our sample, there are not enough articles detailing the results for the *leagues* but there are 12 articles presenting the significance levels and sign associated with the *positions* (Carmichael et al., 1999; Carmichael & Thomas, 1993; Coates & Parshakov, 2022; Franck & Nüesch, 2012; Garcia-del-Barrio & Pujol, 2007; Kalén et al., 2019; Richau et al., 2021; Rubio Martín et al., 2022; Ruijg & van Ophem, 2015; Serna Rodríguez et al., 2019; Speight & Thomas, 1997a, 1997c). Therefore, we can affirm that, in most cases, the reference level is goalkeeper (nine articles) and less often forward (two articles) or midfielder (one article). The nine articles using goalkeeper as the reference level present the results of 20 specifications. At 5%, being a defender is associated with a significantly higher valuation in three specifications, being a midfielder is associated with a significantly higher valuation in four specifications and being a forward is associated with a significantly higher valuation in five specifications. The significant results tend to show that the higher the position on the pitch, the higher the valuation.

4 | DISCUSSION

4.1 | A set of key control variables

Among the variety of football players' valuation drivers that have been identified and tested by several authors over the years, some have become quite common and are today state of the art. In line with previous summarizing work (Frick, 2007; Serna Rodríguez et al., 2019), we show that *age*, usage of the player (*minutes played* or *number of appearances*), *international status* and decisiveness (*goals* and *assists*) positively and consistently influence the valuation of the football player.

Age and *age squared* are the most often tested variables, consistently associated with significant coefficient. We see that, most of the time, *age* has a positive impact on the valuation in the early career of the player and a negative one after some turning point (the quadratic form typically having a negative coefficient). This inverted-U-shape of the relation is explained by the role of two forces moving in opposite directions: experience that grows with age and potential that declines with age. It is assumable that, in early career, the potential of a player should be at its highest and should decrease slowly but faster and faster. In the meantime, the more the player plays, the higher his experience and the higher his valuation. However, some specifications do not significantly test this relation or find other relation (decreasing or U-shaped) and the reasons are numerous: the quadratic form is not included (Adiwiyana & Harymawan, 2021), some covariates confuse the relation (Coates & Parshakov, 2022; Garcia-del-Barrio & Pujol, 2007; Richau et al., 2021), OLS are not used (Dimitropoulos et al., 2018; Majewski, 2016), the sample may be biased toward highly valued players (Franck & Nüesch, 2012; Majewski, 2016; Rubio Martín et al., 2022).

The low-score nature of football can explain why the decisiveness of the players estimated through their ability to score and to assist is positively linked with their valuation. However, the inclusion of one of the usage variables (*minutes played*, *appearances*) in the specifications seems to explain why *goals* and *assists* can be non-significant. Even if the number of specifications in which the performance variables (*goals*, *assists*, *matches*, *minutes played*, *cards*) are found to be significant can seem low, these indicators are often tested conjointly even if they are probably correlated, thus they may weaken each other's influence on the dependent variable.

Also, having played for his *national team*, a player is sending a positive signal to the transfer market, showing his qualities, and attracting lights during national team matches. Even if the variable is non-significant in most of the specifications (14 out of 24), in terms of articles, three found non-significant coefficients while four found significant coefficients. These mixed results might be explained by the heterogeneity of national team qualities. Indeed, the impact may depend on the nationality of the player because playing for one of the best international teams such as France, Germany, or Brazil is associated with high competition. Some studies have thus attempted to take these discrepancies into account by using variables such as *FIFA rank* (Balliauw et al., 2022; Majewski, 2016, 2021; Wicker et al., 2013).

Besides, only focusing on transfer fee regressions allows for the analysis of the buying club characteristics as valuation drivers. The results suggest that the bigger and more successful the buying club, the higher the price of the transferred player. In line with Gerrard and Dobson (2000), we thus consider that building empirical setup without considering clubs' characteristics can be a serious misspecification, particularly when using actual transfer fees as dependent variable. On the other hand, some variables have been tested quite often but are not proven to be significantly impacting the players' valuation. *Yellow cards, red cards, footedness*, and *height* are rarely good valuation predictors.

Also, while football is a global game, the literature tends to focus on leagues from Big-5 countries, and especially on the English ones. Therefore, it is not sure that our results can extend beyond these countries borders but they provide a replicable framework for future research.

Many of the variables that we study in this article could be considered to have mixed results, suggesting that the methods used, the other tested variables, the sample sizes, or the leagues and seasons considered can have a very strong impact on the variable significance levels and signs. Unfortunately, many variations in the way common variables are defined and the inventiveness of researchers to study new variables have prevented us from building a stronger and wider systematic comparison of the specifications results. However, some results stand out even with such discrepancies in empirical settings, thus highlighting their generality. Moreover, we are then able to highlight a number of pitfalls to avoid while working on an econometric model of football players' valuation.

4.2 | Avoiding empirical misspecification

While working on a model of football players' valuation and selecting the control and research variables, scholars should pay specific attention to their empirical setup. First, one of the main empirical challenges is to account for the regulatory framework of football. For example, since the Bosman ruling in 1995 (Union Royale Belges Des Sociétés de Football Association and Others v Bosman and Others, 1995) no transfer fee is claimable at the end of the employment contract of the football player, thus leading the valuation of player to tend toward zero when the contract's expiry date approaches all else being equal. It implies that the time left until contract expiry should be included in empirical designs and that models built to explain or predict post-Bosman transfer fees are missing a critical consequence of the regulation if they do not. It appears that six studies have been conducted on post-Bosman transfer fees (Coates & Parshakov, 2022; Garcia-del-Barrio & Pujol, 2020; Medcalfe, 2008; Richau et al., 2021; Rubio Martín et al., 2022; Ruijg & van Ophem, 2015) and that four of them include the *remaining contract length* as an explanatory variable, all providing evidence that the further the expiry date, the higher the transfer fee (Coates & Parshakov, 2022; Garcia-del-Barrio & Pujol, 2020; Richau et al., 2021; Rubio Martín et al.,

2022). Interestingly, the *total contract length* is also having a positive impact on players' valuation (Garcia-del-Barrio & Pujol, 2020).

More generally, the increased availability of data also brings new challenges to the foreground. With football analytics gaining power, the profusion of performance metrics is reshaping the scouting industry (Wakelam 2022). However, this state of affairs may cause difficulties for scholars working on players' valuation given the potential multicollinearity caused by the abundance of on-the-field performance variables. Fortunately, several recent studies shared extensive correlation matrix allowing us to discuss this issue (Adiwiyana & Harymawan, 2021; Müller et al., 2017; Rubio Martín et al., 2022; Velema, 2019). The first observation to be made is that all absolute performance metrics are dependent on the playing time (either in *minutes* or *games played* as stated above). This suggests that the playing time "neutralizes" other performance variables that, in turn, do not have much explanatory power. Second, it is interesting to see how variables may lessen each other's explanatory power by playing area. For example, on the defensive side, the number of *yellow cards* is correlated with *interceptions* and *fouls* (Müller et al., 2017; Rubio Martín et al., 2022), while on the offensive front, *goals*—already used by early researchers—are strongly correlated with *assists* (Adiwiyana & Harymawan, 2021; Balliauw et al., 2022; Müller et al., 2017; Rubio Martín et al., 2022; Velema, 2019), *fouls won* (Rubio Martín et al., 2022) and *dribbles* (Müller et al., 2017). Given evident multicollinearity issues between performance metrics, it is unsurprising to observe that the empirical results suggest they do not yield any significant explanatory power of players' valuation. As demonstrated above, *goals* and *assists* are strong valuation drivers, but evidence is lacking for other performance indicators.

If on-the-field variables are getting more easily available, on the other hand, contractual data are still missing. When a transfer occurs, several detailed terms of the contracts are rarely disclosed, such as the player wages or potential clauses which may cause measurement errors. We first consider clauses introduced in the employment contract, that is, negotiated between the player and his clubs, and second clauses used in the transfer contract, that is, negotiated between clubs. First, when signing the employment contract, a player and his club can agree beforehand on the compensation fee to be paid to breach the contract. This kind of clause, called a *buyout clause*, therefore constitutes a de facto ceiling for the transfer fee to be paid for that particular player. Also, the club and the player can come to an agreement on diverse bonuses such as a *signing bonus* which is an upfront payment to the player, more frequently offered to players acquired without any transfer fee. At a macro level, the latter could thus encourage players to wait for their employment contract to expire before moving clubs. The generalization of such signing bonus could thus lead to less frequent transfers with fees. Second, regarding the transfer contract signed between clubs, the amount of the fee can be artificially mitigated through contractual mechanisms agreed on. Among these mechanisms we can cite *sell-on clauses* that grant the selling club the right to yield a part of a subsequent potential transfer fee, or *loans with compulsory purchase option* used to postpone the financial impact of the investment in the books. Therefore, valuations could be very ill-measured given the opacity of negotiations and contractual elements. Researchers should at least acknowledge this issue in future work.

Finally, only a minority of articles presenting regressions over players' valuation are backed by strong theoretical framework. Indeed, most of the theories discussed in these articles have not been called upon again in following research pieces to justify the empirical framework. We thus encourage researchers to replicate the results achieved by early researchers and to propose extensions to the theoretical frameworks developed. On top of that, we want to highlight that several variables could have been encompassed in a stronger theoretical framework. First, we can mention that the positive and significant impact of the *youth academy quality* on valuation

(Balliauw et al., 2022), could have been linked to human capital theory. Second, agency theory and information economics could have supported the use of players' agent related variables, even if the *agent's portfolio valuation* have non-significant effect (Herm et al., 2014) and *being represented by an agent* shows mixed evidence (Rubio Martín et al., 2022; Serna Rodríguez et al., 2019). Finally, scholars could have called upon ratchet effect when testing the role of the *N-1 period valuation* as a predictor of the N period valuation, even though the sign of this relationship is not clear (Dimitropoulos et al., 2018; Müller et al., 2017; Velema, 2019; Wicker et al., 2013). We humbly propose these quick "reinterpretations" to show that a theoretical backing of empirical modelling is possible when working with football players' valuation. We hope that this proposal encourages researchers to consider the field as a proper laboratory to test more general economic theories.

4.3 | Limitations

Even if our approach allows to present a big picture of the football player valuation empirical literature, it suffers some limitations. First, we focus on the sampling limitations. In our systematic review approach, we applied some exclusion criteria that prevent us to include articles that may include interesting empirical models. We only included published articles, thus excluding other form of publication with empirical models such as master and PhD thesis, discussion or working papers, or conference papers. Focusing on articles written in English we excluded paper written in other languages. Also, we searched for articles mentioning transfers fees, price, or value but not salaries. Nonetheless, some articles have been using Transfermarkt values as salaries proxies. Thus, if the theoretical background and analysis are not the same, the models used may have been compared with the ones included in our review. Moreover, we consider that transfer fees, Transfermarkt valuation, kicker.de, arbitrated fees, and Marca values are decent proxies to interpret the results of the significance levels and sign analysis. However, the different dependent variables used may lead to biases in the econometric estimations. In a similar way and to have the biggest picture possible, we consider that the several types of models used, and their associated significance levels and sign, are comparable. Refining our analysis to consider only some types of models or only some dependent variables would not have allowed us to have a sufficiently big sample thus making these trade-offs worth it for the sake of the systematic review. Finally, we must acknowledge that the interactions between covariates cannot be considered when we compare the significance levels and signs of the most common variables used in the literature. Hopefully, the limitations of our approach can be overcome by additional future research.

5 | CONCLUSION AND OUTLOOK

Many football stakeholders (players and their agents, clubs and leagues, etc.), as well as football fans, have shown much interest in football players' valuation. Since the early 90s, scholars from various fields have also been keen to investigate the topic and try to uncover the main drivers of this valuation. By conducting a systematic literature review of empirical articles studying football player's valuation to summarize 30 years of empirical findings, this study proposes an overview of the drivers of this valuation and contributes to the literature in four ways. First, it provides the scientific community with a comprehensive list of variables that have been tested in football players' valuation explanatory or predictive models with reference to the articles they appear in Online Appendix C. Second, it classifies these valuation drivers organized into six categories:

Time, Labor, Popularity, Performance, Clubs' characteristics, Players' characteristics (Figure 3) and broken down into subcategories. Third, it highlights a set of routinely tested variables that are good valuation drivers with a high level of evidence (Figures 4 and 5). Fourth, it provides economists with a list of great experiences of well-known theories, for example, Spence's signaling theory, Nash's bargaining theory or superstar economy. Correspondingly, it highlights several theoretical frameworks in which future research focusing on football players' valuation can be embedded. As a result of these contributions, the authors would like to outline the main takeaways from the careful analysis of the literature and offer recommendations for future work.

We encourage researchers to **refer to the economic theory** when setting up their empirical research. In line with that recommendation, empirical settings should also **account for the existing regulatory framework** in order to avoid misspecification. Nowadays, the players' contractual information (salary, contract length) and clubs' characteristics (finance, stadium capacity) are getting more and more easily available and gathered on online platforms, thanks to official press releases and journalists leaks, therefore they should be included in future empirical analysis for they are backed by theory and regulation. Moreover, we suggest that future research **describes the independent variables with regard to the classification proposed** hereabove to streamline the future results comparison. More precisely, when working with performance metrics, we recommend that further research **acknowledges the evident multicollinearity issue** between them and do not force their use for a seemingly limited explanatory power. Furthermore, the findings brought to light **a set of advisable control variables to include**: the *age of the player* at the end of the season, the *number of goals* and *assists* during the last season pre-transfer, the *number of matches* or *minutes played* during the last season and the i*nternational status* of the player. All are expected to positively impact the valuation with the exception of the age which reduces valuation after some tipping point. Including **fixed effects also improve the design** of empirical research, notably the *position* of the player on the pitch and *clubs-related fixed effects*. On the contrary, even if the variables *yellow cards, red cards*, or *footedness* are widely and easily available, their use of in further research is probably not recommended given their repeated lack of significance. Finally, from a practical and technical point of view, it would make sense that researchers **follow similar standards for reporting the significance of their results** (at 1%, 5%, and 10% thresholds).

One of the main contributions of a review is also to propose a research agenda to scholars of the field. Twenty years ago, additional research was requested upon the age-performance relationship and the weight of player age and experience in the equations (Gerrard, 2001). If this goal can be considered relatively achieved, our results and the conclusion of the papers surveyed give a lot of ideas for future research in the field of empirical football player valuation. To clearly depict the possible ways of improvement for the field, we define six objectives:

- **Survey the existing research**. The present study is one of the few attempts to sum up a large and heterogenous field of research in football economics. We think that it can be enhanced and deepened in many ways (see Section 4.3) such as (1) comparing the significance levels and signs in the literature depending on the valuation definition considered, mainly transfer fee and Transfermarkt value; (2) developing the use of systematic reviews and meta-analysis. For example, a meta-analysis on the relation between *age*, the most included explanatory variable, and valuation is of great interest to better understand what is predominant between experience and potential and to found robust age peaks depending on other covariates.

- **Develop the literature on crowdsourced valuation methods**. In this article we assumed that transfer fees and Transfermarkt values are good proxies just as most researchers do. Given the increased number of models developed in the last decade, it seems important to strengthen validity of crowdsourced estimates and to guarantee that they remain good proxies of the transfer fees in the future years.
- **Enhance data availability and spread**. We clearly observed that the wider data availability allowed by Transfermarkt, among others, led to a huge increase in the number of articles focusing on players' valuation. We believe that actions to make data more easily available to all researchers would raise the quantity and improve the quality of articles published. Football data, in particular salaries and transfer fees, are probably very valuable for football stakeholders but every initiative to provide and share open data should be welcomed as it has been done in the performance analysis field (Pappalardo et al., 2019). Most of the time, diverse automatic or optional add-ons proposed to players and clubs as well as sell-on fees are not available either.
- **Use more advanced modeling techniques**. The constant progress in data analysis and machine learning along with the increased computation capacity open new horizons to go beyond ordinary least squares. Some interesting articles were excluded from our research because they used methods delicately interpretable and hardly comparable with other articles. Among the interesting methodologies that future research could develop we can cite partial least squares structural equation modeling (Hofmann et al., 2021), neural networks (Inan & Cavas, 2021), principal component analysis and clustering before MM-regression (Kirschstein & Liebscher, 2018), decision trees and random forests (Richau et al., 2019; Singh & Lamba, 2019; Yigit et al., 2020), penalized regression (Singh & Lamba, 2019; Yigit et al., 2020), or Bayesian model averaging (Serna Rodríguez et al., 2019).
- **Extend the analysis perimeter to out-of-Big-5 leagues**. Some interesting and talent-providing leagues such as the Portuguese Liga or Dutch Eredivisie have been tested by only one article (Velema, 2019). If the attractivity of the big countries' leagues cannot be denied, the importance of the "peripheral" leagues for the transfer market should be assessed and their players' valuation studied. For example, SL Benfica, Porto FC, or Ajax Amsterdam are very important actors of the football player training and trading but they have been rarely included in the studied samples.
- **Test other variable relationship with players' value**. The careful analysis of effects already tested in the literature unravel its gaps and relationships that could be further tested. For example, *total contract length, remaining contract length, youth academy,* or *popularity* (press and social media) have strong theoretical justifications and promising empirical results. Many of these leads are of great interest for football stakeholders in that they can better explain the transfer market functioning, such as the role of agents, clubs' investors, or youth academies in players' career path and valuation. On top of these emerging research questions, some researchers have also proposed some future research ideas to be implemented such as the examination of business and marketing motivations behind decisions on football player transfers (Garcia-del-Barrio & Pujol, 2020), the development of time series (Müller et al., 2017) or the analysis of the interactions between agents and club owners and their impact on players' valuation (Richau et al., 2021).

Looking at the publication years of the selected papers we see that the football players' valuation has been a fast-moving field in the recent years. Therefore, some of these research trends have already been under examination after our last search date, such as the validity of crowdsourced

valuation methods (Campa, 2022), the development of advanced modeling techniques (Aydemir et al., 2022) or the strengthening of evidence on contractual information and valuation link (Poli et al., 2022).

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DATA AVAILABILITY STATEMENT

The data that support the findings of this study are openly available in (Franceschi et al., 2023) at https://zenodo.org/record/7589484.

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ENDNOTES

[1] 2019 is the last registered season free of Covid consequences.

[2] Selection bias is defined by Heckman (1979) as "the bias that results from using nonrandomly selected samples to estimate behavioral relationships," Here, the sample of transferred players is nonrandom.

[3] Mobility to a lower-ranked team is associated with a significant and negative effect on player value (Velema 2019).

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